

- 28** The chlorofluorocarbon CFC-113, $\text{CCl}_2\text{FCClF}_2$, now banned by the international Montreal Protocol, can be converted into CH_2FCF_3 , a less harmful refrigerant, in a three-stage process.

Which stages involve an isomerisation?

- 1** $\text{CCl}_2\text{FCClF}_2 \rightarrow \text{CCl}_3\text{CF}_3$
- 2** $2\text{CCl}_3\text{CF}_3 \rightarrow \text{CCl}_2\text{FCF}_3 + \text{CCl}_3\text{CClF}_2$
- 3** $\text{CCl}_2\text{FCF}_3 + 2\text{H}_2 \rightarrow \text{CH}_2\text{FCF}_3 + 2\text{HCl}$